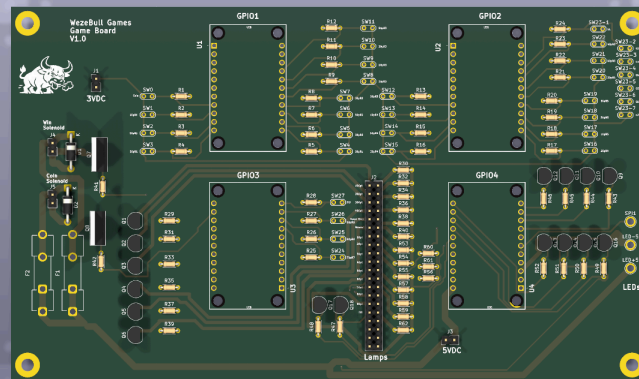




ASSEMBLY GUIDE

SKILLGAME

A TRIBUTE TO THE GAME OF SKILL



ALL HUFF · NO BULL

WezeBull Games · build it once, build it right

Parts list

EVERYTHING YOU NEED BEFORE YOU START

QTY	PART	WHAT IT'S FOR
THE BOARD		
4	Adafruit FT232H breakout	USB→GPIO controllers (U1–U4)
16	PN2222A transistor (TO-92)	lamp / reel drivers
2	TIP120 Darlington (TO-220)	solenoid drivers (Q7/Q8)
2	1N4004 diode	solenoid flyback (D1/D2)
28	10 kΩ resistor	switch pull-downs
16	1 kΩ resistor	transistor base
16	0 Ω jumper (or solder bridge)	lamp links R30–R62 — the #555 bulb self-limits
2	2.2 kΩ resistor	TIP120 base
3	5×20 mm fuses · 4 Keystone clips + 1 inline holder	F1/F2 T2A (solenoids) · F3 T3.5A inline (5 V lead to J3)
1	2×20 pin header (J2)	lamp / output connector
4	2-pin headers (J1/J3/J4/J5)	power in + solenoid out
~34	Playfield switches	28 inputs (S23 = 7 gobble holes)
1	WS2812b LED strip, 150 px	attract / distract shows
20	#555 wedge LED bulb (6.3 V, non-polar)	16 lamps + a 2nd bulb on 100/200/300/400 (double bright)
20	#555 / T10 wedge socket (twist-lock)	one per bulb; solder its leads to the J2 lamp lines
2	SparkFun Solenoid - 5V (Small)	coin-lock + winner-lock
THE SYSTEM		
1	PELADN Ryzen 5 5600H mini-PC (16 GB / 512 GB)	runs SkillGame
1	QQU 15.6" portable touchscreen (with stand)	the display · 1080P IPS/HDR touch
1	4-port USB 2.0 hub	one port per FT232H
1	3.3 V wall adapter	switch / GPIO logic (J1)
1	5 V wall adapter (≥5 A)	lamps · LEDs · coils (J3 / LED±)

TOOLS

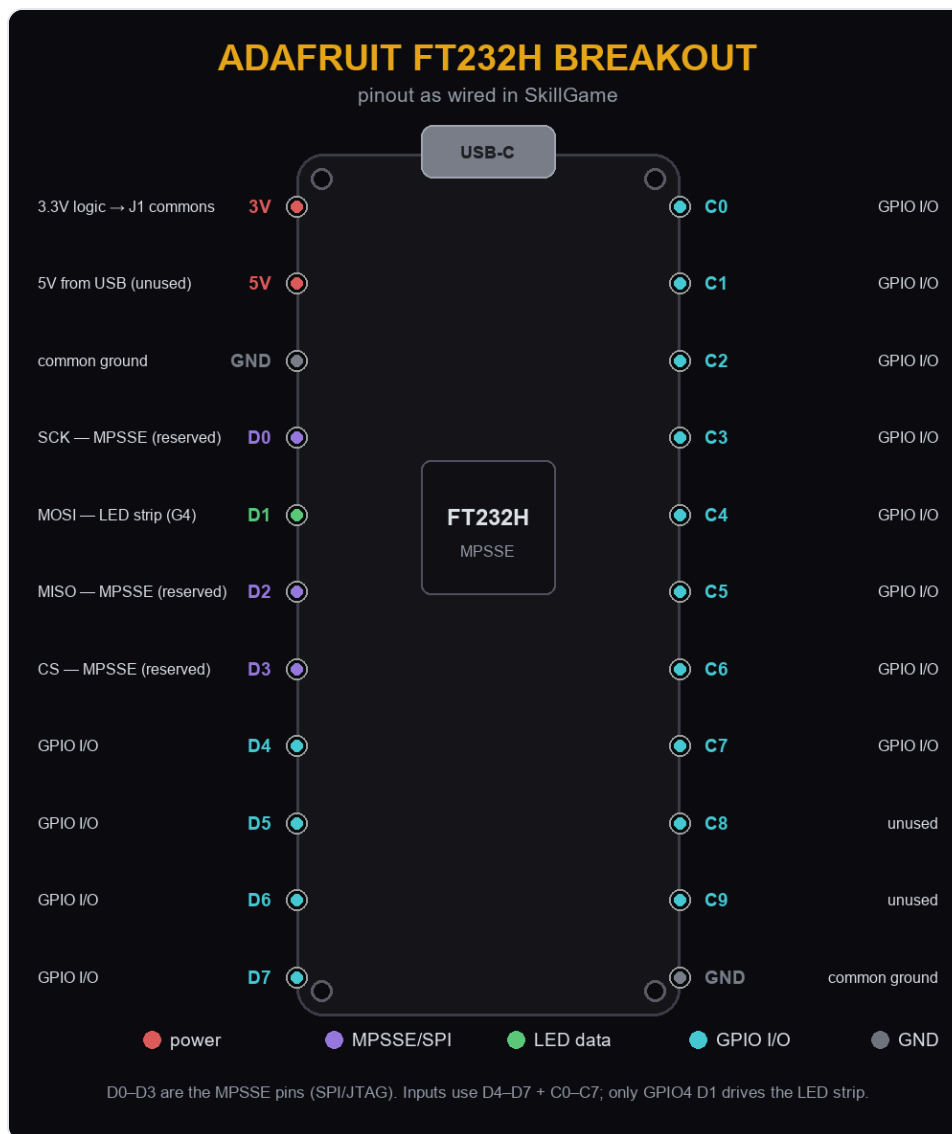
- Soldering iron + solder
- Flush cutters
- Small screwdriver
- Multimeter
- Wire + crimp/ferrules

Before you power up: the 3.3 V adapter goes to J1 (switch commons) and the 5 V to J3 / LED±. Never feed 5 V into J1 — the FT232H inputs are not 5 V-tolerant.

1

Meet the four boards

Adafruit FT232H — one per GPIO bank

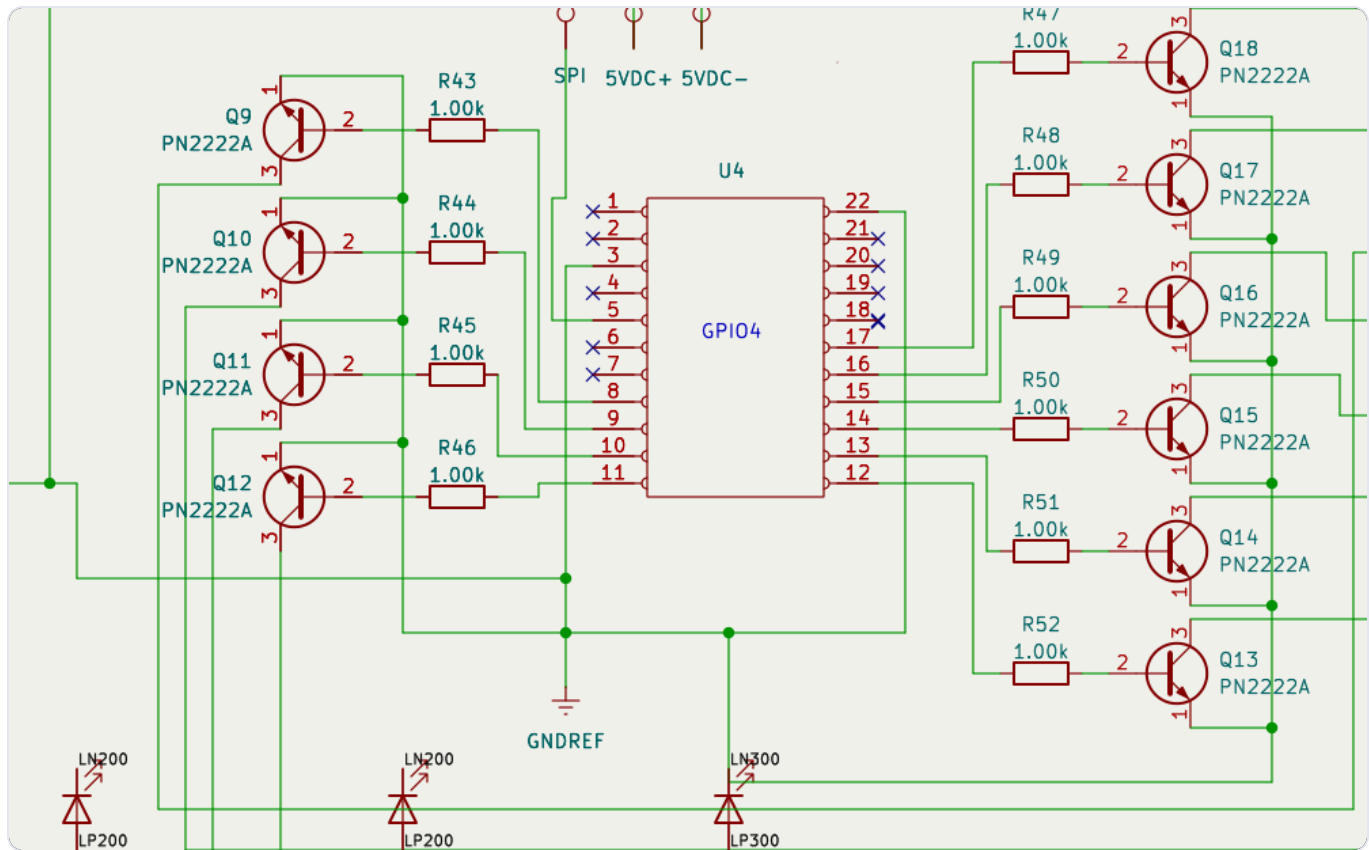


- Each board runs in MPSSE mode over USB-C. D0-D3 are reserved (SPI/JTAG) — only GPIO4's D1 drives the LED strip.
- Program each with its serial (GPIO1-GPIO4) so the software always binds the right board to the right job.

2

Solder the drivers

resistors, then transistors — lowest parts first

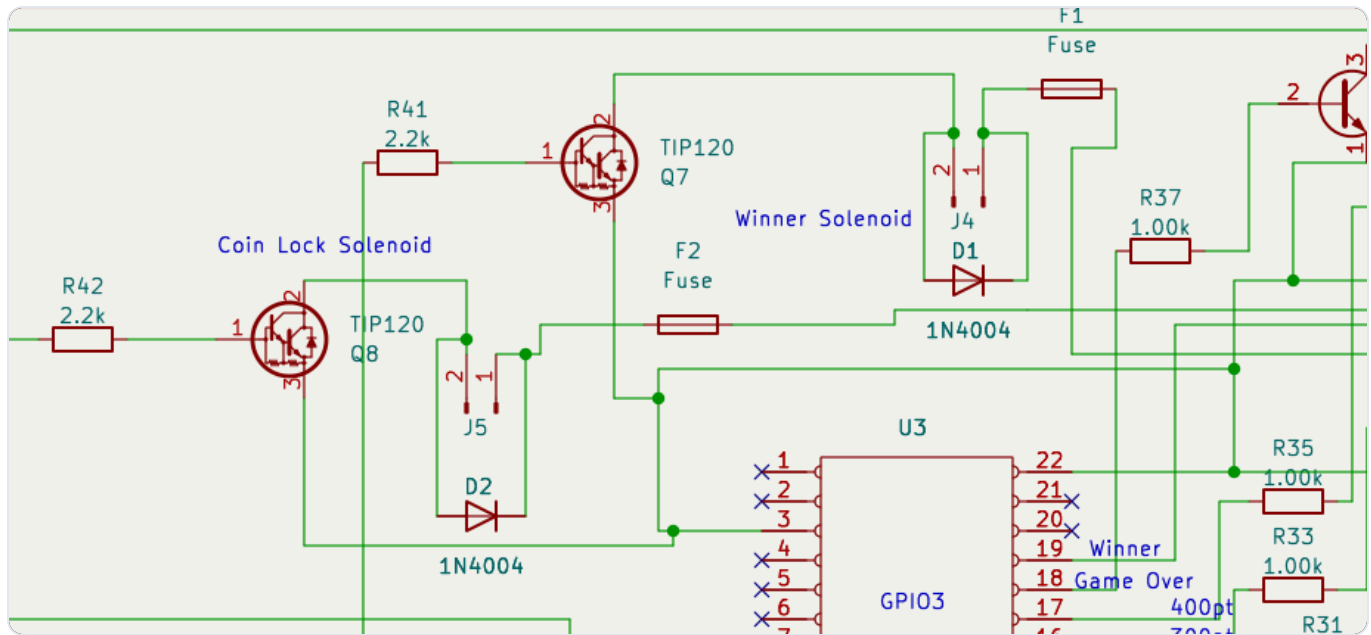


- Resistors first (they lie flat): 28× 10 kΩ pull-downs, 16× 1 kΩ bases, 2× 2.2 kΩ TIP120 bases. The 16 lamp positions (R30–R62) are **0 Ω links** — a jumper wire or solder bridge, since the #555 bulbs self-limit.
- Then the 16 PN2222A transistors (TO-92) — mind the flat/rounded face matches the silk.
- Keep leads short; flush-cut the backs.

3

Solenoid drivers + fuses

TIP120s, flyback diodes, F1/F2



⚠ **TIP120 orientation:** collector goes to the coil, emitter to ground. If you are using an **older board revision**, cross the collector (middle/tab) and emitter (right) leads on Q7 & Q8 — a straight insert wires them backwards and the coils won't switch. The current KiCad files already have this fixed.

- Fit the two TIP120s (TO-220) and the 1N4004 flyback diodes — band matches the silk.
- Install the Keystone 3517 clips and drop in the fuses (see next page).



Fuse it

F1 / F2 — 5 V solenoid feed

FUSES

F1 · F2 solenoids · F3 main 5 V rail

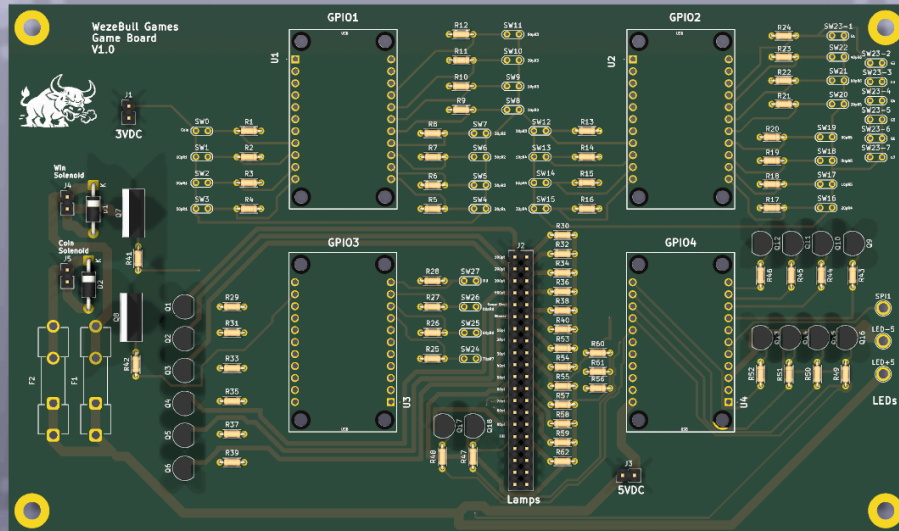
Ref	Protects	In line with	Rating	Part number	Size
F1	Winner-Lock solenoid	+5 V feed to J4	T2A slow-blow 250 V	Littelfuse 0218002.MXP	5 x 20 mm
F2	Coin-Lock solenoid	+5 V feed to J5	T2A slow-blow 250 V	Littelfuse 0218002.MXP	5 x 20 mm
F3	Main 5 V rail (inline)	5 V supply lead into J3	T3.5A slow-blow 250 V	Bussmann GDC-3.5A	5 x 20 mm

Holders: F1/F2 use Keystone 3517 board clips (2 each); F3 uses an inline 5 x 20 mm holder in the 5 V supply lead.
F1/F2 protect the two small 5 V solenoids (SparkFun 'Solenoid - 5V Small') - T2A slow-blow covers their inrush.
F3 guards the whole shared 5 V rail (lamps + LED strip + coils) right where the 5 V adapter feeds J3.
Slow-blow (T) rides out coil / LED inrush; any 250 V 5 x 20 mm part clears 5 V fine.

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Headers & sockets

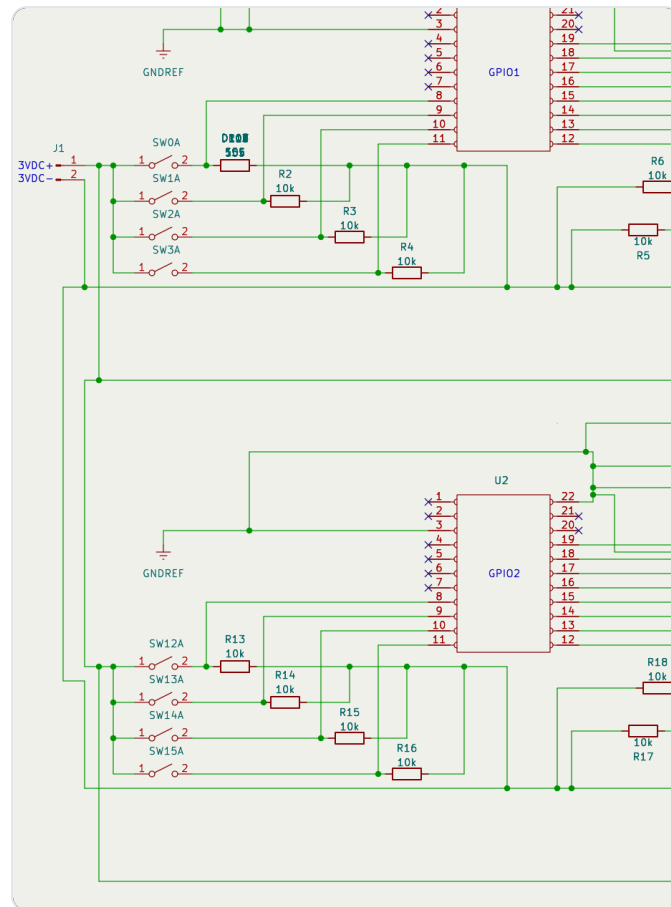
J1–J5, the J2 lamp header, and the 4 FT232H sockets



- Solder the 2×20 header (J2) and the four 2-pin headers (J1/J3/J4/J5).
- Fit the four FT232H sockets. Keep even spacing so the boards sit flat and don't touch — the footprints are laid out with clearance; don't skew them.

Wire the switches

28 inputs → J1 commons, pulled to ground

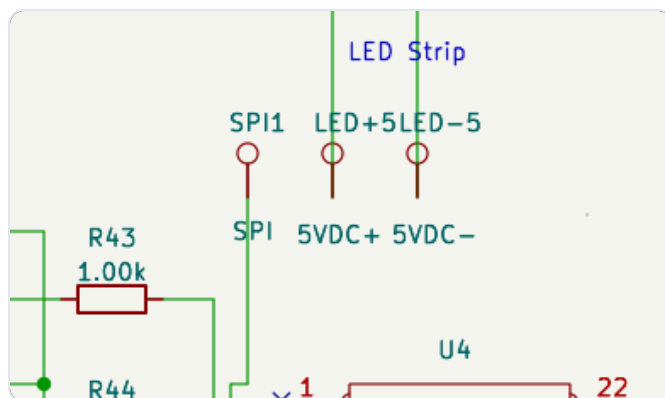
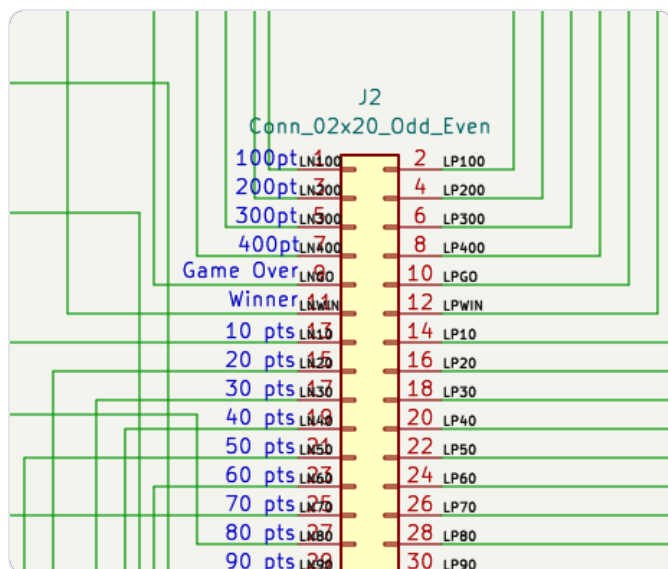


- Every switch closes to the J1 3.3 V common; a 10 kΩ pull-down holds it LOW until then.
- The 7 gobble holes all land on one switch (S23) — wire them in parallel.

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Lamps, coils & LED strip

outputs off J2 / J4 / J5 / SPI



- Each score lamp + Winner/Game-Over/Tilt is an off-board **#555 wedge LED bulb** in a twist-lock socket, wired across its J2 pair (odd pin = switched by the transistor, even pin = +5 V through the 0 Ω link).
- 100 / 200 / 300 / 400** each get **two** bulbs in parallel on the same output — double brightness on the big holes.
- The #555 bulbs are **non-polar**, so socket orientation doesn't matter.
- Coin-Lock → J5, Winner-Lock → J4. LED strip data → GPIO4 SPI (D1); power the strip from its own 5 V.

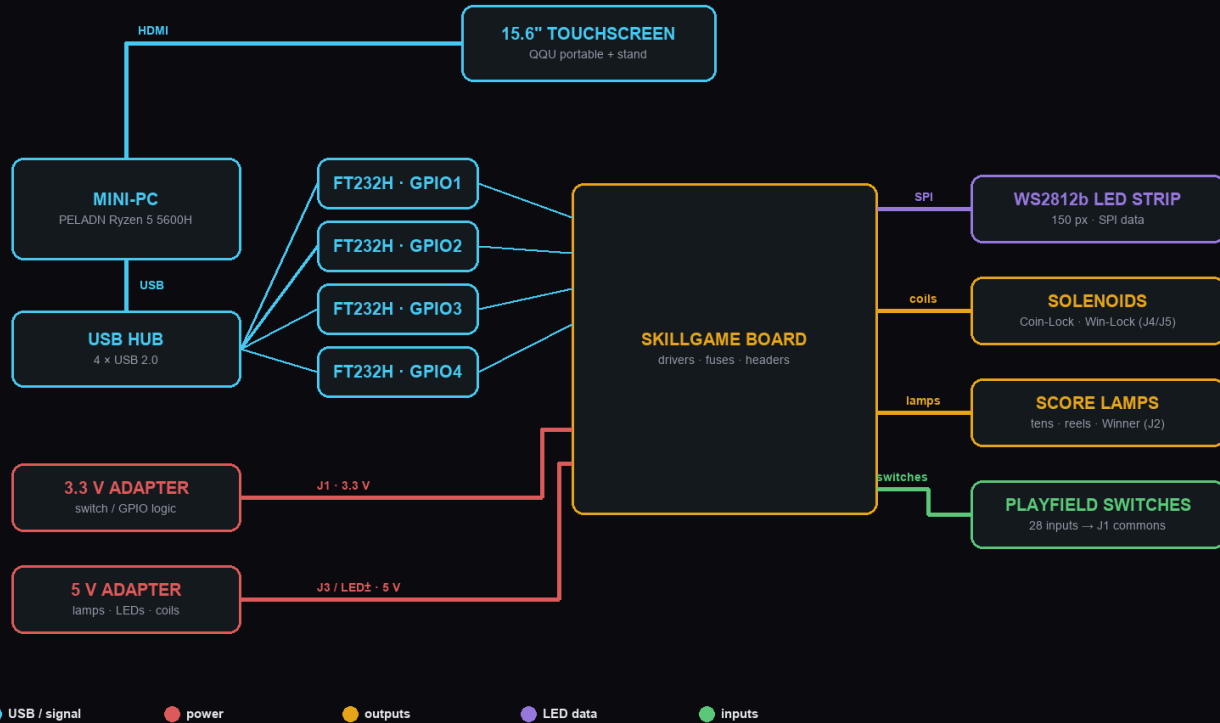
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Hook up the system

PC · hub · monitor · power

SYSTEM HOOKUP

how the pieces physically connect — power, USB and signal



- Mini-PC → HDMI → monitor; Mini-PC → USB hub → the four FT232H boards.
- 3.3 V adapter → J1; 5 V adapter → J3 and LED±. Common all grounds.

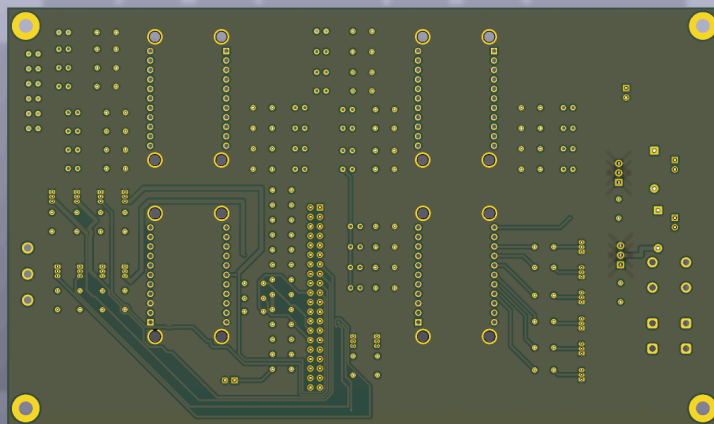
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Power on & self-test

let the software check your wiring

- Launch SkillGame, open **Diagnostics**, and run **Self-Test** — it lights every lamp, pulses the coils and sweeps the LED strip.
- Use the **Switch Monitor** to press each playfield switch and confirm it registers (and clears).
- Any input stuck HIGH shows as a fault ring — check that switch's wiring and pull-down.
- Green badge on Diagnostics = all boards present and healthy. You're done. **ALL HUFF · NO BULL.**

The board back is on the next page for reference, followed by the full schematic.



REFERENCE

